

Problem 7

If $(x + iy) = (u + iv)^n$, where n is an integer and $x, y, u, v \in R$, then $x^2 + y^2 = (u^2 + v^2)^n$. Prove this.

Solution

We need to prove that if $z_1 = u + iv$ is raised to the n -th power to get $z_2 = x + iy$, then the modulus relationship $|z_2| = |z_1|^n$ holds.

Step 1: State What We're Given

Given:

$$x + iy = (u + iv)^n$$

where $n \in Z$ and $x, y, u, v \in R$.

To prove:

$$x^2 + y^2 = (u^2 + v^2)^n$$

Step 2: Recall the Modulus Property

For any complex number $z = a + ib$, the modulus is:

$$|z| = \sqrt{a^2 + b^2}$$

And therefore:

$$|z|^2 = a^2 + b^2$$

Step 3: Apply the Modulus Property

For $z_1 = u + iv$:

$$|z_1|^2 = u^2 + v^2$$

For $z_2 = x + iy$:

$$|z_2|^2 = x^2 + y^2$$

Step 4: Use the Fundamental Property of Modulus

We know from the properties of complex numbers that:

$$|z_1 \cdot z_2| = |z_1| \cdot |z_2|$$

More generally, for any complex number z and integer n :

$$|z^n| = |z|^n$$

Step 5: Apply to Our Problem

Since $x + iy = (u + iv)^n$, we have:

$$|x + iy| = |(u + iv)^n|$$

Using the property from Step 4:

$$|x + iy| = |u + iv|^n$$

Step 6: Square Both Sides

Squaring both sides:

$$|x + iy|^2 = (|u + iv|^n)^2 = |u + iv|^{2n}$$

But we can also write:

$$\begin{aligned} |x + iy|^2 &= x^2 + y^2 \\ |u + iv|^{2n} &= (|u + iv|^2)^n = (u^2 + v^2)^n \end{aligned}$$

Therefore:

$$x^2 + y^2 = (u^2 + v^2)^n$$

This completes the proof.

Alternative Proof: Direct Calculation

For Positive Integer n

We can prove $|z^n| = |z|^n$ by induction.

Base case: $n = 1$: $|z^1| = |z|^1$

Inductive step: Assume $|z^k| = |z|^k$. Then:

$$|z^{k+1}| = |z^k \cdot z| = |z^k| \cdot |z| = |z|^k \cdot |z| = |z|^{k+1}$$

So by induction, $|z^n| = |z|^n$ for all positive integers n .

For Negative Integer n

If $n < 0$, let $m = -n > 0$. Then:

$$z^n = z^{-m} = \frac{1}{z^m}$$

$$|z^n| = \left| \frac{1}{z^m} \right| = \frac{1}{|z^m|} = \frac{1}{|z|^m} = |z|^{-m} = |z|^n$$

So the property holds for negative integers as well.

For $n = 0$

$$z^0 = 1 \implies |z^0| = |1| = 1 = |z|^0$$

(assuming $z \neq 0$)

Step 7: Verification with Example

Let $u = 1$, $v = 1$, and $n = 2$.

$$(u + iv)^2 = (1 + i)^2 = 1 + 2i + i^2 = 1 + 2i - 1 = 2i$$

So $x = 0$, $y = 2$.

Check:

$$x^2 + y^2 = 0^2 + 2^2 = 4$$

$$(u^2 + v^2)^n = (1^2 + 1^2)^2 = 2^2 = 4$$

Indeed, $x^2 + y^2 = (u^2 + v^2)^n$.

Step 8: Geometric Interpretation

In the complex plane:

- $|u + iv| = \sqrt{u^2 + v^2}$ is the distance from origin to point (u, v)
- $|x + iy| = \sqrt{x^2 + y^2}$ is the distance from origin to point (x, y)
- The equation states: When raising a complex number to the n -th power, the modulus gets raised to the n -th power
- This is consistent with the polar form: $(re^{i\theta})^n = r^n e^{in\theta}$

Connection to Polar Form

In polar form:

$$u + iv = r(\cos \theta + i \sin \theta) = re^{i\theta}$$

where $r = \sqrt{u^2 + v^2}$.

Then:

$$(u + iv)^n = r^n(\cos(n\theta) + i \sin(n\theta)) = r^n e^{in\theta}$$

The modulus is:

$$|(u + iv)^n| = r^n = (\sqrt{u^2 + v^2})^n = (u^2 + v^2)^{n/2}$$

And:

$$|(u + iv)^n|^2 = (u^2 + v^2)^n$$

which equals $x^2 + y^2$.

Final Answer

$$\boxed{x^2 + y^2 = (u^2 + v^2)^n}$$

Proof: Using the fundamental property $|z^n| = |z|^n$ for any complex number z and integer n :
From $(x + iy) = (u + iv)^n$, taking modulus of both sides:

$$|x + iy| = |(u + iv)^n| = |u + iv|^n$$

Squaring both sides:

$$x^2 + y^2 = (u^2 + v^2)^n$$

Key principle: The modulus of a power equals the power of the modulus: $|z^n| = |z|^n$.
This holds for all integers n (positive, negative, or zero), provided $z \neq 0$ when $n \leq 0$.